

Lem prod combinacion convexa prod finito blaschke

Lema 1. Sean f y g combinaciones convexas de productos finitos de Blaschke. Entonces, $f \cdot g$ es una combinación convexa de productos finitos de Blaschke.

Demostración: Por definición, sabemos que $\exists \{f_k\}_{k=1}^n, \{g_\ell\}_{\ell=1}^m$ productos finitos de Blaschke y $\{\alpha_k\}_{k=1}^n, \{\beta_\ell\}_{\ell=1}^m \subset \mathbb{R}_{\geq 0} : \sum_k \alpha_k = 1 \wedge \sum_\ell \beta_\ell = 1 \wedge f = \sum_k \alpha_k f_k \wedge g = \sum_\ell \beta_\ell g_\ell$. Luego,

$$f \cdot g = \left(\sum_{k=1}^n \alpha_k f_k \right) \cdot \left(\sum_{\ell=1}^m \beta_\ell g_\ell \right) = \sum_{k=1}^n \sum_{\ell=1}^m \alpha_k \beta_\ell f_k g_\ell.$$

Como f_k y g_ℓ son productos finitos de Blaschke, $f_k g_\ell$ es un producto finito de Blaschke. Además, $\sum_k \sum_\ell \alpha_k \beta_\ell = (\sum_k \alpha_k)(\sum_\ell \beta_\ell) = 1$. Por lo tanto, $f \cdot g$ es una combinación convexa de productos finitos de Blaschke. ■

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